# Topic 5 Mean Field Theory and Phase Transitions

## Review of entire class so far!

### Large deviation theory

- N independent samples $\vec{s}$
- from true distribution $p(\vec{s})$
- $\hat{p}$ = empirical distrib
- at large N:

$$\text{Prob}[\hat{p}] \approx e^{-N D_{KL}[\hat{p} \| p]}$$

### KL Divergence

$$kT\, D_{KL}[q \| p] = G(q) - F(\beta)$$

### Open physical system

microstates $\vec{s}$

energies $E(\vec{s})$

in thermal eq at temperature $kT = \frac{1}{\beta}$

higher energy states exp less likely because reservoir has exp fewer states

$\beta$ = growth rate of reservoir entropy $S$ w/ reservoir energy $E$

$$\beta = \frac{1}{kT} = \frac{\partial S}{\partial E}$$

### Gibbs Free Energy

$q(\vec{s})$ = any other distribution

$$G(q) \equiv \langle E \rangle_q - kT\, S(q)$$

$$S(q) \equiv -\sum_{\vec{s}} q(\vec{s}) \ln q(\vec{s})$$

$$p(\vec{s}) = \arg\min_q G(q)$$

$$F(\beta) = \min_q G(q)$$

### Boltzmann distribution:

$$p(\vec{s}) \equiv \frac{1}{Z(\beta)} e^{-\beta E(\vec{s})}$$

### Partition function:

$$Z(\beta) \equiv \int d\vec{s}\, e^{-\beta E(\vec{s})}$$

### (Helmholtz) Free energy:

$$F(\beta) = -\frac{1}{\beta} \ln Z(\beta)$$

### Maximum Entropy

$q(\vec{s})$ is any distribution

$$S(q) = -\sum_{\vec{s}} q(\vec{s}) \ln q(\vec{s})$$

$$p(\vec{s}) = \arg\max S(q)$$

subject to $\langle E \rangle_q = E_0$

$\beta(E_0)$ Lagrange multiplier

$$F(\beta) = \langle E \rangle_p - kT S(p)$$

### Free Energy is a Cumulant generating func for energy

### Tilted free energy as cumulant gen func for

### Macroscopic Limits, Saddle Points and Legendre Transform

$$C_n(E) = (-1)^n \frac{\partial^n}{\partial \beta^n}\left[-\beta F/\beta\right]$$

other observables

$$E(\vec{s},h) = E(\vec{s}) - kT\, h\,\theta(\vec{s})$$

$$C_n(\theta) = \frac{\partial^n}{\partial h^n}\left[-\beta F(\beta,h)\right]\Big|_{h=0}$$

if

$$E = N\varepsilon \qquad S(E) = N s(\varepsilon)$$

$$\varepsilon,\ s(\varepsilon) \sim O(1)$$

as $N \to \infty$

then

$$f(\beta) = \varepsilon^* - kT\, s(\varepsilon^*)$$

$$\varepsilon^* = \operatorname{argmin}\left[\varepsilon - kT\, s(\varepsilon)\right]$$

$$\beta = \frac{\partial S}{\partial E}$$

or $\quad \beta f(\beta) + s(\varepsilon^*) = \beta \varepsilon^*$

$$\beta \leftrightarrow \varepsilon$$

$$\beta f(\beta) \leftrightarrow s(\varepsilon)$$

Legendre dual

Fluctuation-Response Theorems

$$\frac{\partial \langle E \rangle_p}{\partial \beta} = -\mathrm{Var}(E)$$

$$\frac{\partial \langle \theta \rangle_p}{\partial h} = \mathrm{Var}(\theta)$$

Large deviation principle for energy density $\varepsilon = E/N$ at large $N$

$$p(\varepsilon) \simeq e^{-N I(\varepsilon)}$$

$$I(\varepsilon) = \left[\varepsilon - kT\, s(\varepsilon)\right] - f(\beta)$$

Done with general formalism : now examples

# Simplest example: A single spin

$\uparrow \quad s = +1 \qquad\qquad E = -hs$

$\downarrow \quad s = -h$

$$Z(\beta, h) = \sum_{s=\pm 1} e^{+\beta h s} = e^{\beta h} + e^{-\beta h} = 2\cosh \beta h$$

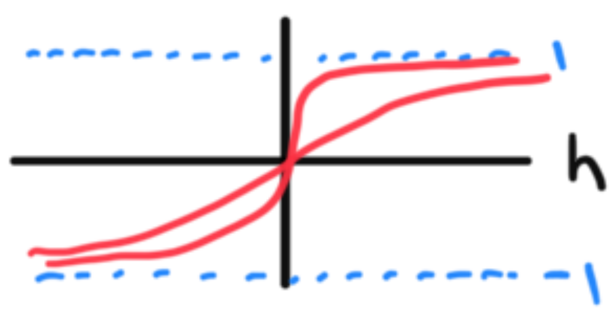
$$\cosh x = \frac{e^{x} + e^{-x}}{2}$$

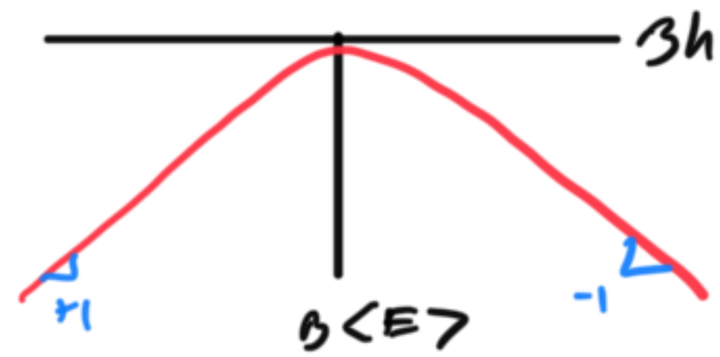
$$\boxed{F(\beta, h) = -\frac{1}{\beta} \ln 2\cosh(\beta h)}$$

$$\langle E \rangle = -\frac{\partial}{\partial \beta}(\beta F) = \frac{\partial}{\partial \beta}\left[\ln 2\cosh(\beta h)\right]$$

$$= h \, \frac{2\sinh(\beta h)}{2\cosh(\beta h)} = \boxed{h \tanh(\beta h) = \langle E \rangle}$$

$$\langle E \rangle = h\langle s \rangle \Rightarrow \boxed{\langle s \rangle = \tanh(\beta h)}$$

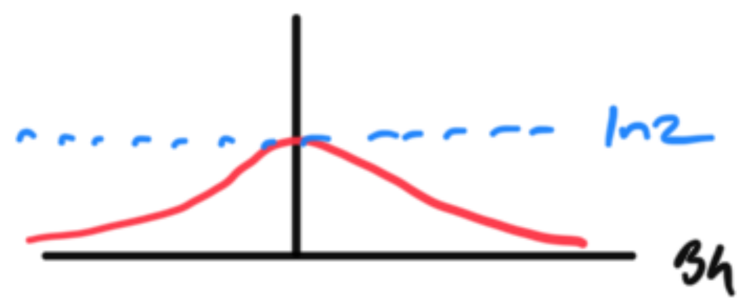



$$F = \langle E \rangle - \frac{1}{\beta}\langle S \rangle$$

$$\boxed{S(\beta) = \beta\left[\langle E \rangle - F(\beta)\right]}$$

$$= \beta\left[\ln 2\cosh(\beta h) + \beta h \tanh \beta h\right]$$




## Example: our first interacting system Mean field ferromagnet

$$E(\vec{s}) = -\frac{1}{N}\sum_{i,j=1}^{N} J\, s_i s_j - h\sum_{i=1}^{N} s_i$$

$\underline{h=0}$

$+J$ ↑↓ ↓↑

$-J$ ↑↑ ↓↓

set $J=1$

let $$m = \frac{1}{N}\sum_{i=1}^{N} s_i$$ sample mean magnetization

note $$m^2 = \left[\frac{1}{N}\sum_{i=1}^{N} s_i\right]\left[\frac{1}{N}\sum_{j=1}^{N} s_j\right]$$

$$= \frac{1}{N^2}\sum_{i,j=1}^{N} s_i s_j$$

so $$E(\vec{s}) = -Nm^2 - Nhm$$

energy of microstate $\vec{s}$ depends on $\vec{s}$ only through macroscopic variable $m$ !!

$$P(\vec{s}) \propto e^{-\beta E(\vec{s})} \Rightarrow P(m) \propto e^{-\beta E(m)}\, e^{S(m)}$$

# microstates with magnetization $m$ ...

if $pN$ spins are up and $(1-p)N$ spins are down $\Big\}$ $m = p - (1-p) = 2p-1 = m$

$$p = \frac{m+1}{2}$$

so # of configurations w/ magnetization $m$ = # of configs with frac $p = \frac{m+1}{2}$ spins up

we know this is $e^{NS(p)}$ with $s(p) = -p\ln p - (1-p)\ln(1-p)$

$$p = \frac{1+m}{2}$$

$$1-p = \frac{1-m}{2}$$

$$s(m) = -\frac{1+m}{2}\ln\frac{1+m}{2} - \frac{1-m}{2}\ln\frac{1-m}{2}$$

<u>Summary:</u>

$$P(m) \propto e^{-\beta\left[\underbrace{E(m) - kTNS(m)}_{F(m)\ \text{restricted free energy}}\right]}$$

$$P(m) \propto e^{-N\beta\left[\underbrace{(-m^2 - hm) - kT\,s(m)}_{f(m)\ \text{free energy density}}\right]}$$

$$\boxed{P(m) \approx e^{-N I(m)}}$$

large deviation principle for $m$

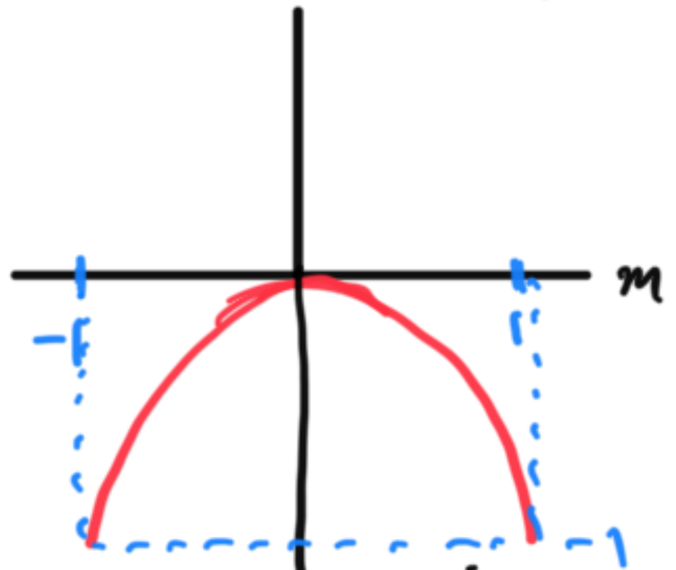
$$\boxed{\begin{aligned} I(m) = \;& \beta\left[\mathcal{E}(m) - kT\,S(m)\right] \\ & - \beta\left[\mathcal{E}(m^*) - kT\,S(m^*)\right] \end{aligned}}$$

$$f(m) = \mathcal{E}(m) - kT\, s(m)$$

$$\mathcal{E}(m) = -m^2 - h\, m$$

when $h=0$

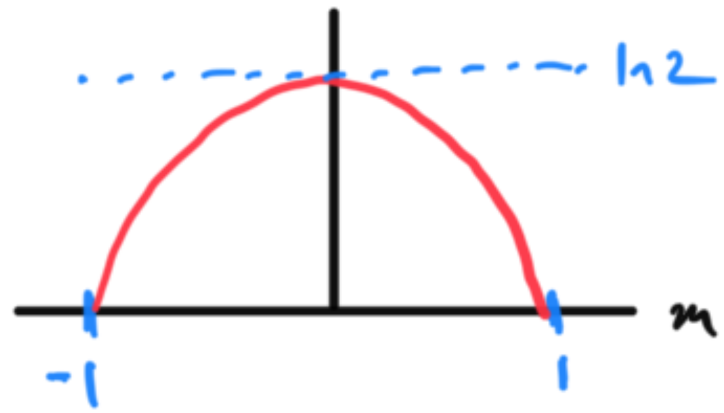
$\mathcal{E}(m)$ w/ $h=0$




Minimize energy $\mathcal{E}(m)$

send $m \to \pm 1$

$s(m)$




Maximize entropy $s(m)$

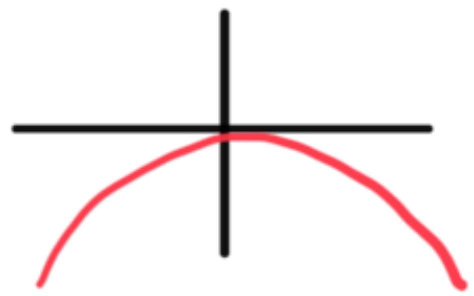
send $m \to 0$

the most likely value of $m$

is $\boxed{m^* = \arg\min f(m)}$

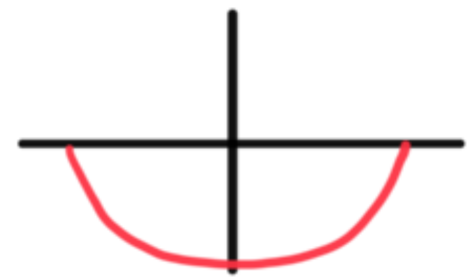
arises as a competition between

$\mathcal{E}(m)$

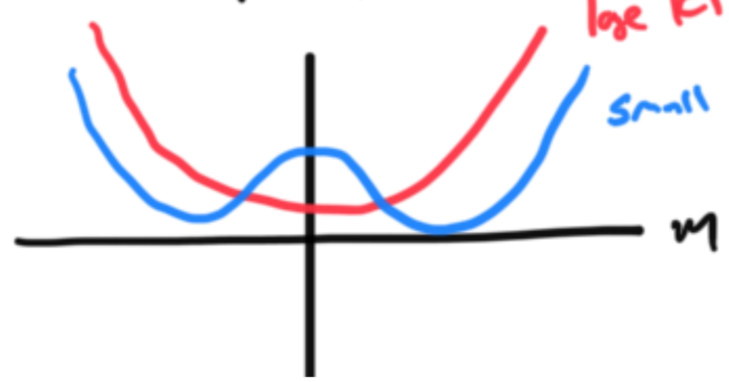


$+ \; kT$

$-s(m)$



$f(m)$

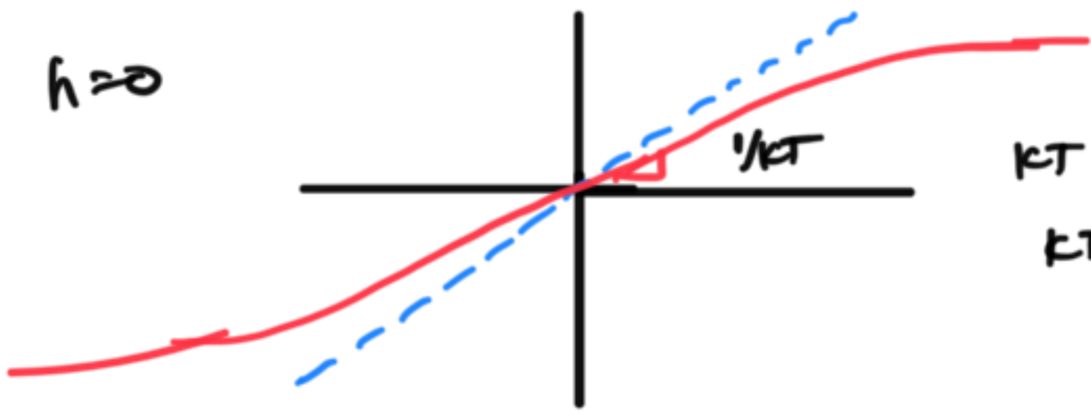


$$\frac{\partial f}{\partial m} = 0 \quad \Rightarrow \quad \frac{1}{2}\log\frac{1+m}{1-m} = \beta(m+h)$$

$$\tanh^{-1} x = \frac{1}{2}\log\frac{1+x}{1-x}$$

$$\boxed{m = \tanh \beta(m+h)}$$

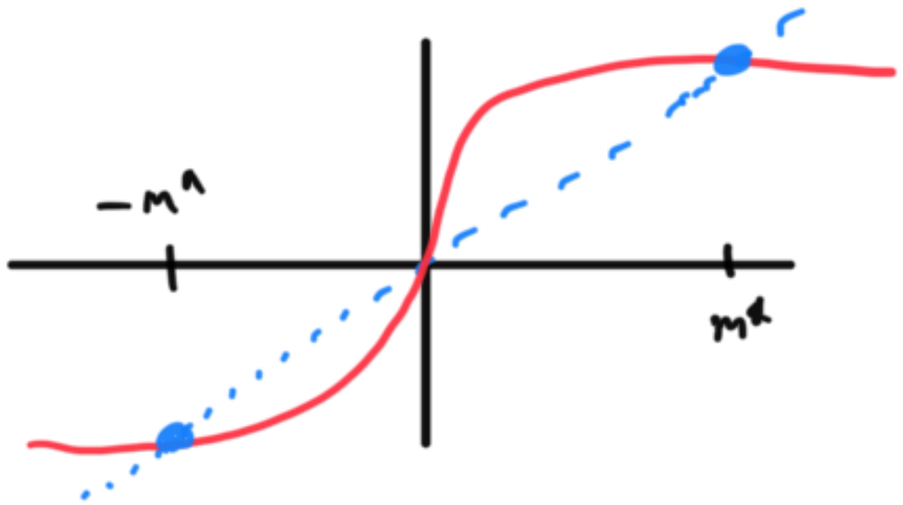
h=0




$m = \tanh m/KT$

KT large

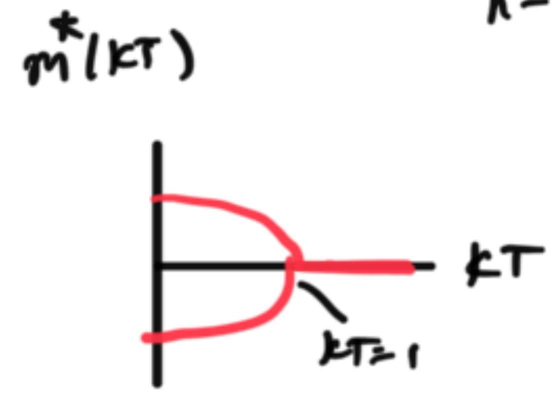
$KT > 1 \qquad 1/KT < 1$




KT small

$KT < 1 \qquad 1/KT > 1$

h=0




2nd order / continuous phase transition

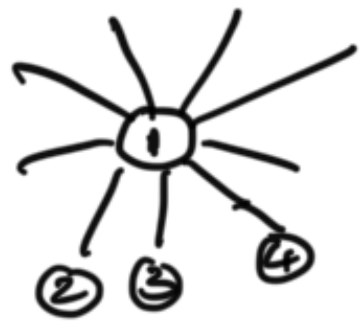
$m^*(\beta)$ continuous

$\frac{\partial m^*}{\partial \beta}$ diverges

$\sim$ 2nd deriv of free energy diverges

## Heuristic derivation

$$E(\vec{s}) = -\frac{1}{2N}\sum_{i,j} J s_i s_j - h\sum_i s_i$$




· assume each spin has magnetization $m = \langle \sigma_i \rangle$

- Consider a single spin $s_1$

- all terms in energy function involving $s_1$:

$$E(s_1) = -\sigma_1 \left[ \frac{1}{N} \sum_{j=1}^{N} J s_j + h \right]$$

now $s_j = \underbrace{m}_{\text{mean}} + \underbrace{s_j - m}_{\text{fluctuating}}$

$$E(s_1) = \underbrace{-\sigma_1 \left[ J m + h \right]}_{\text{mean field applied to } \sigma_1} \underbrace{- J\sigma_1 \left[ \sum_{j=1}^{N} (s_j - m) \right]}_{\text{neglect fluctuating}}$$


$\longrightarrow$ $h^{MF} = Jm + h \longrightarrow s_1$

Interacting system $\longrightarrow$ single spin in a mean field

self-consistency:

$$m = \langle s_1 \rangle = \tanh \beta h^{MF}$$

$$\boxed{m = \tanh \beta (Jm + h)}$$

## Variational Mean field derivation

assume a product form $q(\vec{s}) = \prod_{i=1}^{N} b(s_i)$

$b(s_i)$ is a distribution with $\langle s_i \rangle_q = m$

$s(b) = S(m)$ above

$S(q) = N\, S(m)$

then
$$\langle E \rangle_q = \left\langle -\frac{J}{N} \sum_{ij} s_i s_j - h \sum_j s_j \right\rangle_q$$

$$= -\frac{J}{N} \sum_{ij} \langle s_i s_j \rangle_q - h \sum_j \langle s_j \rangle_b$$

$$= -\frac{J}{N} \sum_{ij} \langle s_i \rangle_b \langle s_j \rangle_b - h \sum_j \langle s_j \rangle_b$$

$$= -\frac{J}{N} \sum_{ij} m^2 - h \sum_j m$$

$$= N \Big[ \underbrace{-J m^2 - h m}_{= \mathcal{E}(m)} \Big]$$

then $G(q) = \langle E \rangle_q - kT\, S(q)$

$$G(q) = N \left[ \mathcal{E}(m) - kT\, S(m) \right]$$

$$\min_q G(q) \longleftrightarrow \min_m \left[ \mathcal{E}(m) - kT\, S(m) \right]$$

# First order discontinuous transition in Potts model

$\vec{s}$ can now take $k$-states ie $s_i \in \{1, 2, 3, \dots q\}$

$$H = -J \sum_{i,j} \delta_{s_i, s_j} \qquad \delta_{s_i,s_j} = \begin{cases} 1 & \text{if } s_i = s_j \\ 0 & \text{otherwise} \end{cases}$$

variational mean field : $q(\vec{s}) = \prod_{i=1}^{N} b(s_i)$ $\qquad b = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_K \end{bmatrix} \qquad b(s_i = j) \equiv f_j$

<u>$q = 3$</u>

$$G(q) = -KNJ\left[f_1^2 + f_2^2 + f_3^2\right] + NKT\left[\sum_j f_j \ln f_j\right]$$

$f_i \geq 0$

$f_1 + f_2 + f_3 = 1$